Pedro **Juan Royo** CURRICULUM VITAE

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PhD student in the Day group at the University of Southampton. Highly independent worker, and up to tackle any challenge.

Experience

The University of Sheffield, Department of Materials Science & Engineering **Sheffield, UK**
Intern *June 2020 - July 2020*

- Analyzed the impedance response of $BaTiO_3$ under different conditions using **COMSOL**. Found interesting relations between the aspect ratio of the grains of the material with its impedance response.
- Developed **MATLAB** code to analyse the results and create visualisations.
- This work was presented at the **WorldCUR-BCUR 2023 Conference** in the University of Warwick.

Institut Català de Nanociència i Nanotecnologia (ICN2) **Barcelona, Catalonia**
Intern *July 2019 - August 2019*

- Created **core-shell silica-polystyrene nanoparticles** to study the effects of nano-plastics on living tissue.
- Learned how to **independently work on a lab.**

Skills

- **Languages**: Catalan (Native), Spanish (Native), English (Professional working skills), French (Speaking and writing skills).
- **Tools**: Version control software (Git), LaTeX, Typst, SQLite, Ansys (limited), Solid Edge (limited)
- **Coding**: Python, Rust

Education

- **PhD student** / University of Southampton / 2024 - Now
  - Investigating the potential energy surfaces of molecular crystals in relation to crystal structure prediction workflows. Mostly using the Monte Carlo threshold clustering algorithm.
  - Developed the skills to carry out independent work, and manage a project.
- **MEng Materials Science and Engineering (Research stream)** / The University of Sheffield / 2020 - 2024

Achievements

Mappin Medal and Premium Prize *July 2024* **Sheffield, UK**

- Prize is awarded to ten undergraduate students annually, in honour of the late Sir Frederick Mappin, Bart. It is given to students who have shown great distinction, and are in the final year of study for an undergraduate programme offered by the Faculty of Engineering.

CiMs-Cellex Scholarship *May 2018* **Barcelona, Catalonia**

- **Obtained one of the 24 positions of the competeitive CiMs-Cellex scholarship** to course International Baccalaureate in Aula Escola Europea.

HP Codewars Sant Cugat Competition *2017* **Sant Cugat, Catalonia**

- **Ranked 1st** in a coding competition organised by HP.

Projects

marimol Self-motivated *August 2026*

- Python package to visualize molecular structures in marimo notebooks, providing a clean interface and reactivity for downstream cells.

mol-CSPy Day group code *2024 - Now*

- Actively developed the code in my research group in Southampton.
- Lead group meetings to ensure development of the code progresses adequately and with reasonable quality.